ClipSAT : Cambridge & Edexcel Additional Mathematics 0606/4PM1

Unit 7.01: Differentiation & Rates of Change

Name: _____

Q1. Find the equation of the tangent line to the curve y = 2x + 1^3 at the point where x = 0 .

[ ] A: y = 2x + 1
[ ] B: y = 6x - 1
[ ] C: y = 6x + 1
[ ] D: y = 6x + 3

Q2. Find the equation of the tangent line to the curve y = 3x + 1^3 at the point where x = 0 .

[ ] A: y = 9x + 3
[ ] B: y = 9x - 1
[ ] C: y = 3x + 1
[ ] D: y = 9x + 1

Q3. Find the equation of the tangent line to the curve y = 4x + 1^3 at the point where x = 0 .

[ ] A: y = 12x + 3
[ ] B: y = 12x - 1
[ ] C: y = 4x + 1
[ ] D: y = 12x + 1

Q4. Find the equation of the tangent line to the curve y = 5x + 1^3 at the point where x = 0 .

[ ] A: y = 15x + 3
[ ] B: y = 15x - 1
[ ] C: y = 5x + 1
[ ] D: y = 15x + 1

Q5. Find the equation of the tangent line to the curve y = 6x + 1^3 at the point where x = 0 .

[ ] A: y = 18x + 1
[ ] B: y = 6x + 1
[ ] C: y = 18x - 1
[ ] D: y = 18x + 3

Q6. Find the equation of the tangent line to the curve y = 7x + 1^3 at the point where x = 0 .

[ ] A: y = 21x - 1
[ ] B: y = 7x + 1
[ ] C: y = 21x + 3
[ ] D: y = 21x + 1

Q7. Find the equation of the tangent line to the curve y = 8x + 1^3 at the point where x = 0 .

[ ] A: y = 8x + 1
[ ] B: y = 24x + 3
[ ] C: y = 24x - 1
[ ] D: y = 24x + 1

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Q8. Air is pumped into a spherical balloon such that its radius increases at a constant rate of 0.5 text{ cm/s} . What is the rate of change of volume frac{dV}{dt} when the radius is r = 3 text{ cm} ?

[ ] A: 18pi text{ cm}^3/text{s}

[ ] B: 36pi text{ cm}^3/text{s}

[ ] C: 9pi text{ cm}^3/text{s}

[ ] D: frac{18}{2}pi text{ cm}^3/text{s}